

An ABM Model of the College Admissions Process

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Introduction

The higher education system in the United States has often come under criticism for perpetuating economic inequality due to the lofty cost of attendance paired with the increasing necessity of a degree in order to obtain a job. Students from financially stable families who can afford to pay the cost of attendance without a significant amount of loans face less challenges post-college. However, students whose families are not able to pay for college might end up not enrolling in college which can make finding a job difficult in some instances. Students might also borrow detrimental amounts of money to cover the costs of college and later have to pay off their debt which can range from a minor inconvenience to a lifelong struggle. The issue of economic inequality in higher education has made its way to several politicians' platforms. In 2021, Representative Pramila Jayapal (Washington) and Senator Bernie Sanders (Vermont) introduced the College For All Plan to Congress (Csreinicke). The legislation aimed to reduce the economic burden of attending college to achieve "the standard of living the American people deserve" (Csreinicke). Additionally, President Biden has canceled \$127 billion in student debt since taking office in 2021 (Lobosco). However, the cost of attending college may not be the only challenge that students face. Standardized test scores have been linked to the wealth of a student's family as well. The purpose of this model is to examine the mechanisms for economic inequality in higher education, specifically the cost of attendance and the impact of wealth on standardized test scores, in a simplified agent-based setting.

Model Overview

Stage 0: Initialization

Two classes of agents are created in this model, students and colleges. For the simulations, the number of students generated was 10,000 and the number of colleges created was 30. Students' attributes include resources which is an integer between 0 and 100, ACT score which is an integer generated on a normal distribution of mean 20 and standard deviation 6 and clipped to the range 10 to 36 which is based on ACT data, a floating point essay score generated on a normal distribution mean 5 and standard deviation 2.5 clipped to the range 0 to 10, an arbitrary "fit" which is a floating point number between 0 and 10, and a fit preference which is a floating point number between 0 and 1 and represents the weight a student places on fit between them and the college.

The attributes of colleges include a cost which is an integer between 0 and 100, a minimum ACT which is an integer from 14 to 32, an essay weight which is a floating point number between 0 and 1 that encodes how much weight a college puts on a student's essay, the school's test optional status which is a binary quantity (roughly 80% of schools are test optional in this model which reflects the landscape in the United States as of late 2023), a pool of aid that a college can award to students which is an integer from 1,000 to 10,000 and is used to make up the difference between the college's cost and the student's resources, an ideal class size which is an integer from 100 and 600 (based on the ratio of students to colleges in the model), ideal population size which is four times the initial ideal class size, an over offer multiplier which is a floating point number between 1 and 3 and signifies how many more students a college will admit than their ideal class size to account for competition between schools, and an arbitrary "fit" which is a floating point number between 0 and 10. The "fit" attribute is used between a student and a college with the smaller magnitude of the difference between making the student more likely to apply and the student more likely to enroll at the college. Colleges also have attributes to keep track of their specific information including the size of each of the four classes currently enrolled at the college, their acceptance and yield rates (initially set to zero), and the average ACT of students that enroll at the college (initially set to zero). This information is used by students to make decisions about what schools they apply to and updates the ideal class size for a school and their over offer multiplier which are described in more detail under Stage 5.

Stage 1: ACT Wealth Adjustment

After the ACT scores for students are generated as described above, the scores are then adjusted for wealth based on the function below.

$$New\ ACT = ACT + \frac{43}{750}resources + \frac{193}{60}$$

The equation is based on a report about wealth adjusted ACT scores released by the ACT organization (McNeish, Daniel, et al.). This report outlined the average difference in composite scores between students with family incomes below \$36,000, between \$36,000 and \$80,000, and above \$80,000. The average composite score of the students between \$36,000 and \$80,000 was taken as the baseline and set at a resource level of 50. The average difference between the baseline group and the other two groups was set at a resource level of 25 and 75. From this, a line was created to give the ACT adjustment at each resource level, and it was added to the students' ACT score that was generated in the Initialization stage. The results were made an integer and clipped to the range of 10 to 36.

Stage 2: Application

In the Application stage, students are the active agents as they decide the schools to which they will submit applications. The "distance" between the students and the colleges are

calculated using the formula below where ϕ and β are constants. For this model, ϕ was set to 0.5 and β was set to 0.1.

$$\begin{aligned} distance &= \phi e^{-(studentfit - schoolfit)^2} + \beta(1 - acceptancerate) \text{ if } acceptancerate > 0 \\ distance &= \phi e^{-(studentfit - schoolfit)^2} \text{ if } acceptancerate = 0 \end{aligned}$$

The student then loops through each of the 30 colleges and decides whether to apply or not. If the college is not test optional, then the student will apply to the school if the student's ACT score is higher than the school's minimum ACT and with a probability equal to the "distance". If the school is test optional, then the student will apply if their ACT score is higher than ten less of the average ACT score of students that have previously enrolled at the school with a probability equal to the "distance".

Stage 3: Admission

In the Admission stage, colleges are the active agents as they decide what students to admit from their list of applicants. First, the colleges calculate the number of applicants they are going to admit by multiplying their ideal class size by their over offer multiplier. If the school is test optional, they admit the students with the highest essay scores up to their designated size. If the college is not test optional, they loop through their applicant list and put students on the admitted list until they reach their designated size. After they reach their designated size and if they still have students to consider, they take the student with the lowest ACT already on the list and the student with the lowest essay already on the list and compare the attributes of the next student that the college is considering. If the next student that the college is considering has a higher essay and ACT than the students with the minimum values, they are added to the list. The student with the minimum essay on the list is removed with a probability corresponding to the school's essay weight and the student with the minimum ACT on the list is removed with a probability corresponding to one minus the school's essay weight. If the student has a higher essay score than the minimum and a lower ACT than the minimum, then the student is added to the list and the student with the minimum essay score is removed with a probability equal to the school's essay weight (otherwise nothing changes). If the student has a lower essay score than the minimum and a higher ACT than the minimum, then the student is added to the list and the student with the minimum ACT score is removed with a probability equal to one minus the school's essay weight (otherwise nothing changes). If the student has a lower essay score and ACT, they are not added to the list and nothing changes.

Stage 4: Enrollment

In the Enrollment stage, the students are the active agents as they take the list of colleges that accept them and decide which college to attend. First, students go through the list of colleges

to which they were admitted and determine which colleges they can afford to narrow down their options. If the student's resources are greater or equal to the school's cost, then the student adds it to their short list. If the student's resources are less than the school's cost and the difference is less than the financial aid resources of a college, then the student adds the school to their short list. The college's financial aid resources are reduced by the amount needed to make up the difference. The student then determines the school that is the best fit, which is measured by the smallest magnitude of the difference of the student's fit and the college's fit, the school with the highest average ACT of students that have enrolled there before, and the school with the lowest acceptance rate. The student enrolls in the school with the best fit with a probability equal to the student's fit preference and otherwise randomly chooses between the school with the highest average ACT and the school with the lowest acceptance rate.

Stage 5: Updating Information and Iteration

After the process is completed, the colleges update their attributes. Each college updates their total acceptance rate and yield rate. The acceptance rate is used by students to select colleges. The yield rate is used to calculate the over offer multiplier that the school will use next year which is given by the formula $overoffer multiplier = \frac{\theta}{yieldrate}$ where θ is constant that was set to 0.75 in this model. Each school also keeps track of their first year, sophomore, junior, and senior class sizes as well, so these are advanced with the senior class size dropping out and the number of enrolled students becoming the first year class size. One fourth of the difference of the ideal population of the school and the sum of these classes is then added to the ideal class size to determine the ideal class size for the next year. The average ACT of all the students that have enrolled at the school is updated as well. Finally, the financial aid resources of the school are reset for the next year. These stages are then repeated for a specified number of iterations which correspond to yearly application cycles.

Simulations and Results

Simulation 1: No Wealth Impacts

The set up of Simulation 1 included no wealth impact on the ACT and the college cost was set to zero. Financial aid resources of colleges and essay scores of students were also set to zero. The purpose of this simulation was to ensure that the model was working properly, as there should be no trends across different student resource levels, and the enrollment rate of students should be almost entirely based on ACT scores. Below are figures summarizing the results of the simulation.

	<u>1 year</u>	<u>10 years</u>	<u>20 years</u>

% of students enrolled	61.73	60.81	60.73
Avg resources of enrolled students over entire time period (/100)	50.72	50.04	50.09
Avg resources of unenrolled students over entire time period (/100)	49.63	50.05	50.03

Figure 1: Numerical summaries of Simulation 1 after 1 year, 10 years, and 20 years. The model reaches a stable equilibrium quickly as the data after 1 year is roughly equivalent to the data after 20 years. Additionally, the average resources of all enrolled students over the entire time period for each simulation time is roughly equivalent to the average resources of all unenrolled students which are both roughly 50/100, indicating a level of economic equality in the outcomes.

Acceptance Rate of Students by Resources and ACT

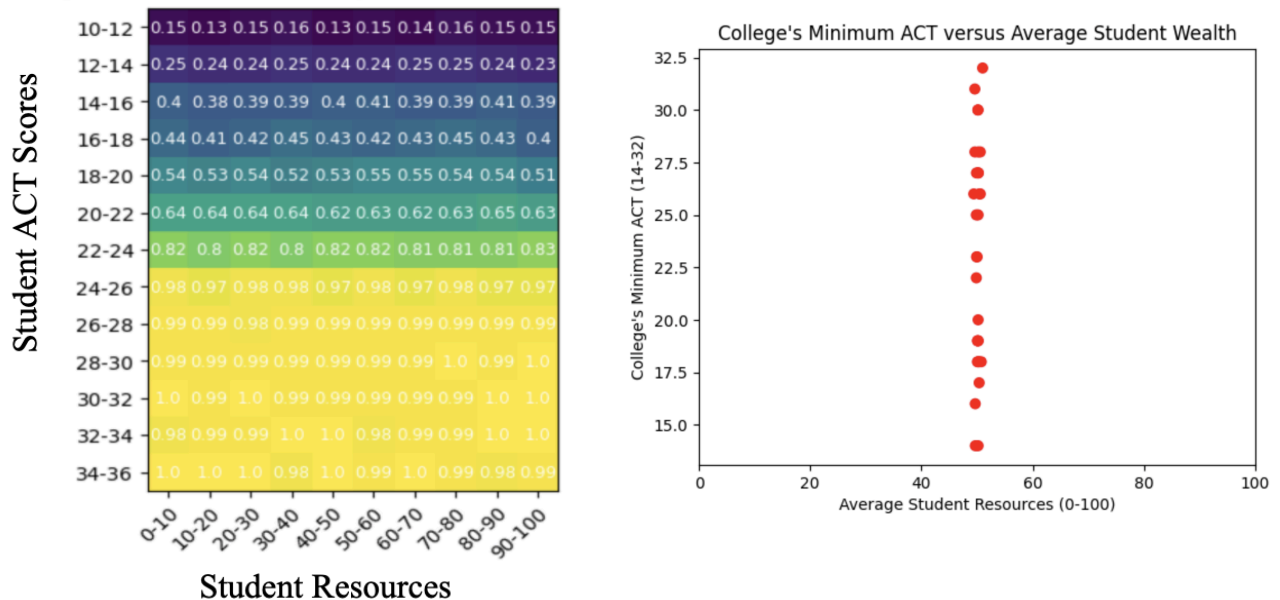


Figure 2: Heatmap of enrollment rate for students by ACT score and resource level over 20 years (left) and colleges' minimum ACT requirements versus average resources of all students enrolled at the college over a 20 year period (right). From these plots, it is apparent that there is no disparity in outcomes due to resource level, only ACT score, and there is a degree of economic equality at each school.

Simulation 2: Add Cost and Wealth Adjusted ACT

Simulation 2 takes the same structure as Simulation 1, but includes a cost for colleges and a wealth adjustment to ACT scores. Below are figures summarizing the outcomes of this simulation.

	<u>1 year</u>	<u>10 years</u>	<u>20 years</u>
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% of students enrolled	40.47	40.02	39.80
Avg resources of enrolled students (/100)	64.79	66.04	66.23
Avg resources of unenrolled students (/100)	40.47	39.31	39.41

Figure 3: Numerical summary of Simulation 2 after 1 year, 10 years, and 20 years. The model reaches a stable equilibrium similarly to Simulation 1. Unlike Simulation 1, a significantly lower percentage of students enroll in college, and there is a significant difference between the average resources of all enrolled students versus the average resources of all unenrolled students which is roughly 27/100 over 20 years.

Acceptance Rate of Students by Resources and ACT

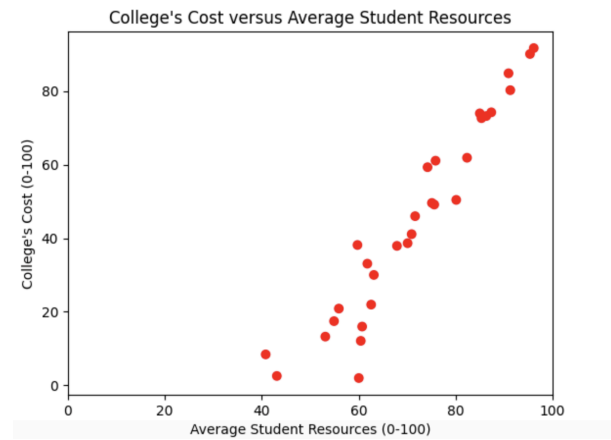
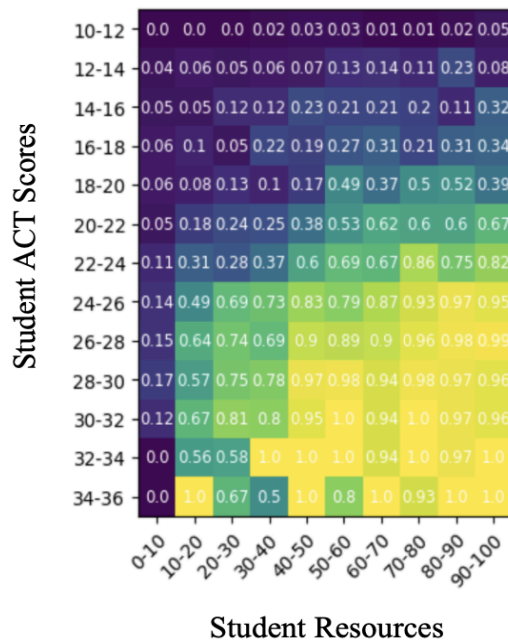


Figure 4: Heatmap of enrollment rate for students by ACT score and resource level over 1 year (left) and colleges' minimum ACT requirements versus average resources of all students enrolled at the college over 1 year (right). The heatmap shows a concentration of enrollment rates in the quadrant of the plot that corresponds to above average resources and above average ACT scores. The outcomes in this region of the plot have stark differences when compared with other regions of the plot. The college's cost versus the average resources of their enrolled students shows a positive correlation. Thus, more expensive schools have wealthier students in this simulation even after just 1 year.

Acceptance Rate of Students by Resources and ACT Scores

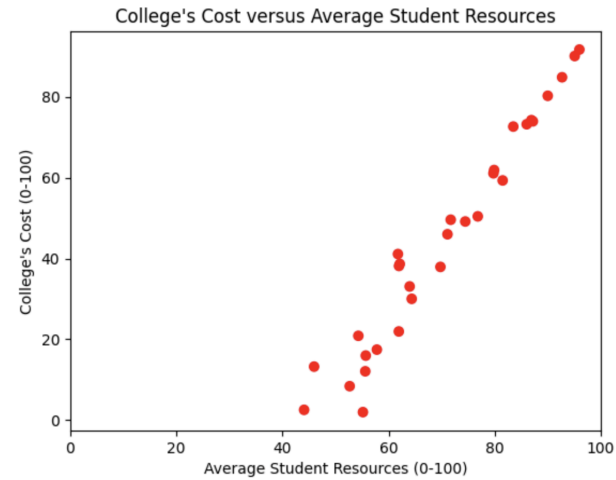
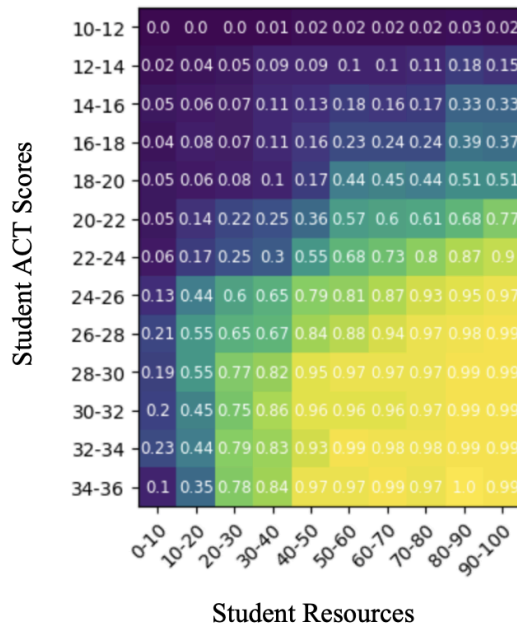


Figure 5: Heatmap of enrollment rate for students by ACT score and resource level over 20 years (left) and colleges' minimum ACT requirements versus average resources of all students enrolled at the college over a 20 year period (right). The heatmap shows a further concentration of enrollment rates in the quadrant of the plot that correspond to above average resources and above average ACT scores compared to Figure 4. The difference in outcomes at different levels are well defined and stark. Additionally, the positive correlation between a college's cost and the average resources of their enrolled students is stronger when compared to Figure 4. Thus, the disparities in outcomes are exacerbated over time.

Simulation 3: Add Financial Aid and Essay

Simulation 3 is built upon Simulation 2 by adding a financial aid mechanism described in the Model Overview section and an essay score for students which was not impacted by wealth. Additionally, roughly 80% of the schools were made test optional which reflects the landscape in higher education in the United States as of 2023 (Churchill). These schools were given an essay weight of 1. In the scatter plots below, test optional schools are represented by blue dots, and schools that were not test optional are represented by red dots. In reality, test optional schools might still accept standardized testing scores, however I used the essay score to encapsulate the facets of an application that are not impacted by wealth, and wondered if only requiring these components would improve economic diversity at schools. The following figures summarize the results of Simulation 3 after 1 year and after 20 years.

	<u>1 year</u>	<u>10 years</u>	<u>20 years</u>
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% of students enrolled	70.89	64.22	57.68
Avg resources of enrolled students (/100)	53.64	54.98	55.00
Avg resources of unenrolled students (/100)	43.24	42.65	44.07

Figure 6: Numerical summary of Simulation 3 after 1 year, 10 years, and 20 years. Based on this data, the model does not reach a stable equilibrium as it did during Simulations 1 and 2 in terms of the percentage of total students enrolled which decreases over time, but does stay relatively stable in terms of the average resources of enrolled and unenrolled students. There is a degree of economic inequality as the average resources of enrolled students is about 11 points higher than the average resources of unenrolled students. However, this difference is significantly less than the disparity in Simulation 2, and a greater percentage of students enroll in college compared to Simulation 2.

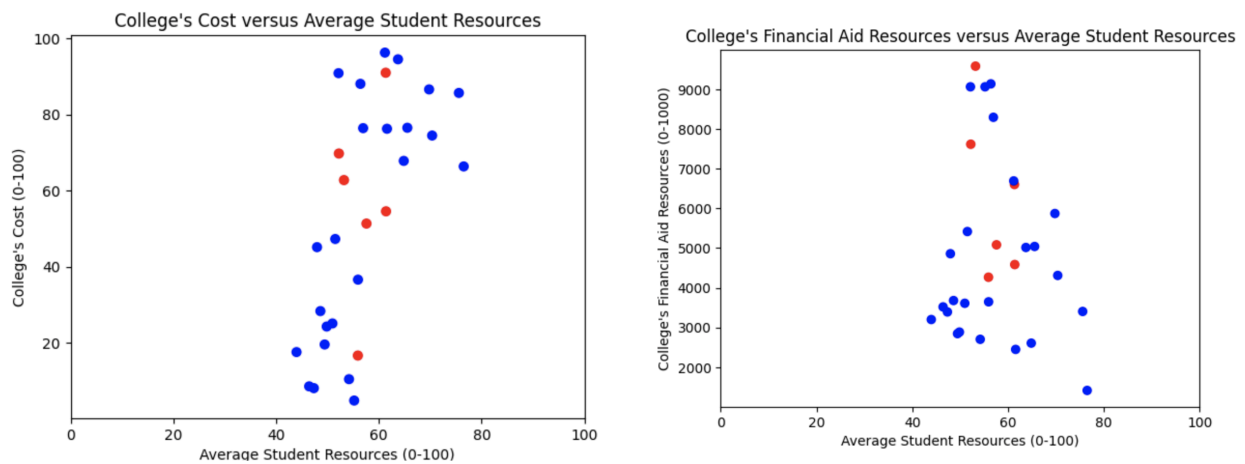
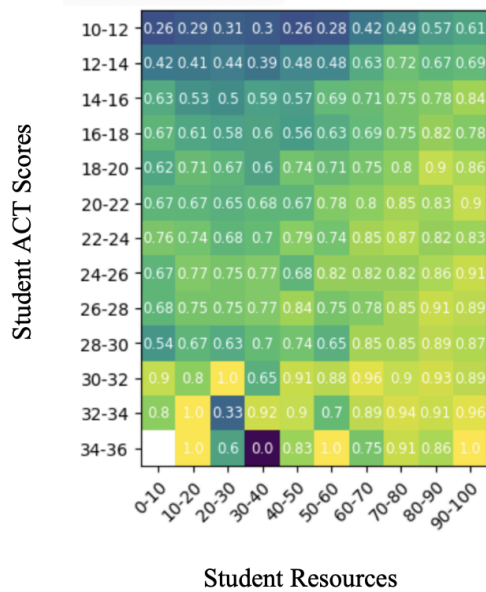


Figure 7: The cost of colleges versus the average resources of their enrolled students after 1 year (left) and college's financial aid resources versus the average resources of their enrolled students after 1 year (right) with blue dots representing test optional schools and red dots representing schools that are not test optional. There is a somewhat strong positive correlation between a college's cost and the average resources of their enrolled students, especially for more expensive colleges, but this correlation is significantly weaker than that seen in Simulation 2. There is not a significant correlation between a college's financial aid and the average resources of their enrolled students, but there is a wider range of average resources at colleges with lower financial aid resources which likely depends on its cost. Neither of these trends appears to be impacted by a school being test optional or not.

Acceptance Rate of Students by Resources and ACT Scores



Acceptance Rate of Students by Resources and Essay Scores

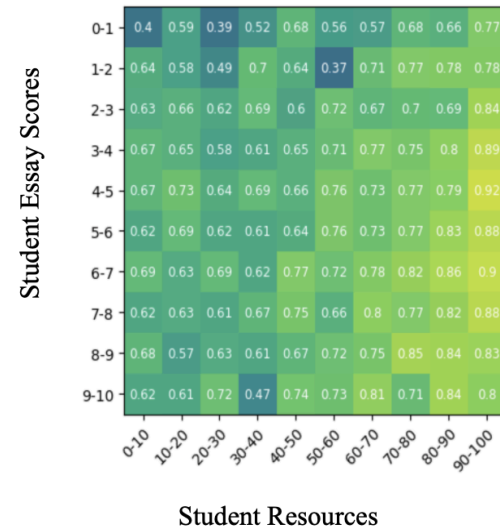


Figure 8: Heatmap of enrollment rate for students by ACT score and student resources (left) and heatmap of enrollment rate for students by essay score and student resources (right) both after 1 year. The heat maps show a less stark difference in student outcomes based on their attributes as seen in Simulation 2. However, there are still higher enrollment rates for wealthier students. It is also important to note that due to the ACT wealth adjustment and the normal curve on which scores are generated, very few students have ACT scores above 30 and a resource level below 30, so each student's outcome has a bigger impact on the heatmap than in other regions. This can create deceptively high enrollment rates in certain areas of the heatmap which can be rectified by iterating this process over a number of years.

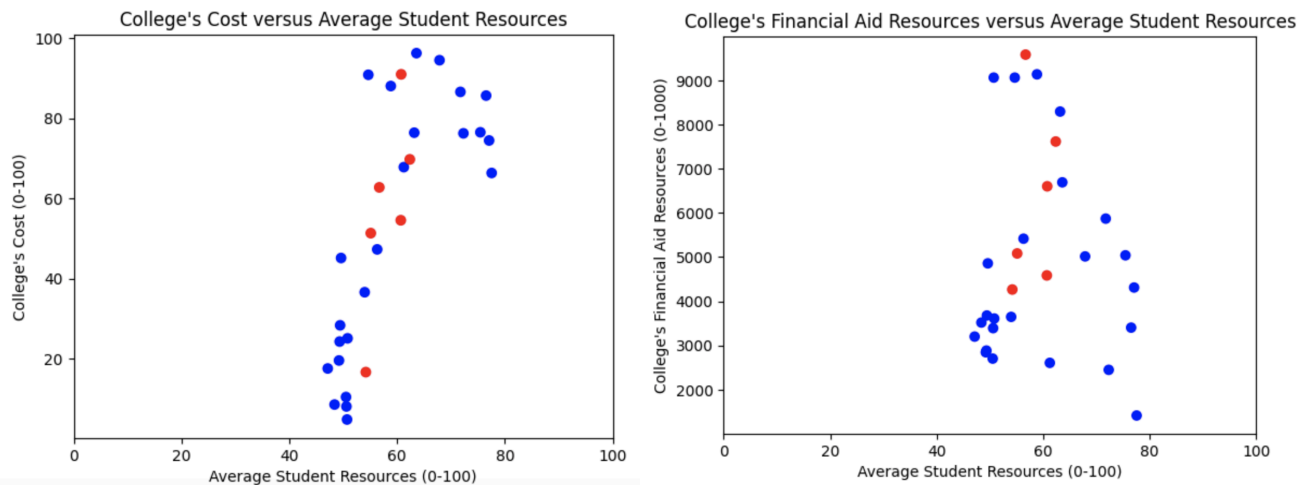
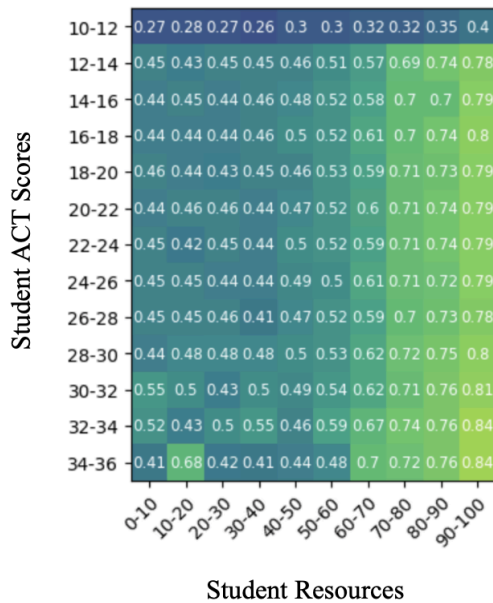


Figure 9: The cost of colleges versus the average resources of their enrolled students after 20 years (left) and colleges' financial aid resources versus the average resources of their enrolled

students after 20 years (right). There is a somewhat strong positive correlation between a college's cost and the average resources of their enrolled students, especially for more expensive colleges, and this correlation is stronger than that seen in Figure 7. The financial aid resources of a college has a somewhat negative correlation with the average resources of their enrolled students which makes sense as schools that have less resources to help students who are unable to pay to attend will end up admitting wealthier students.

Acceptance Rate of Students by Resources and ACT Scores



Acceptance Rate of Students by Resources and Essay Scores

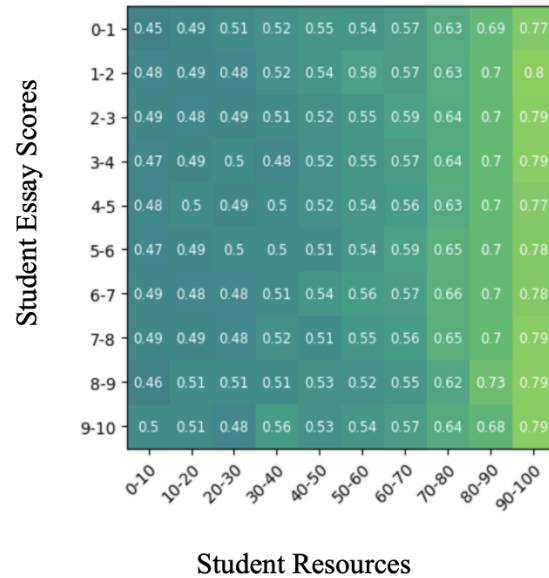
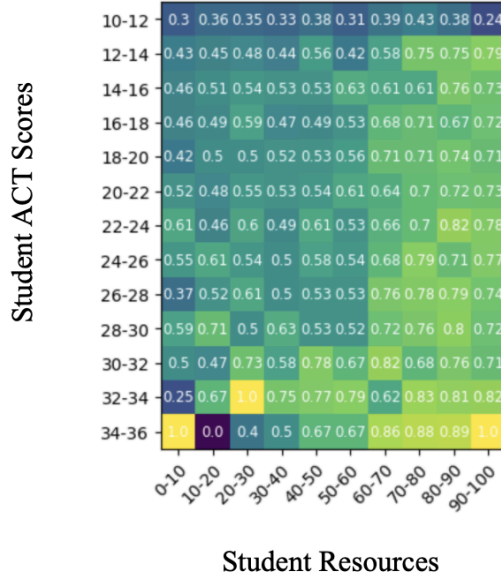


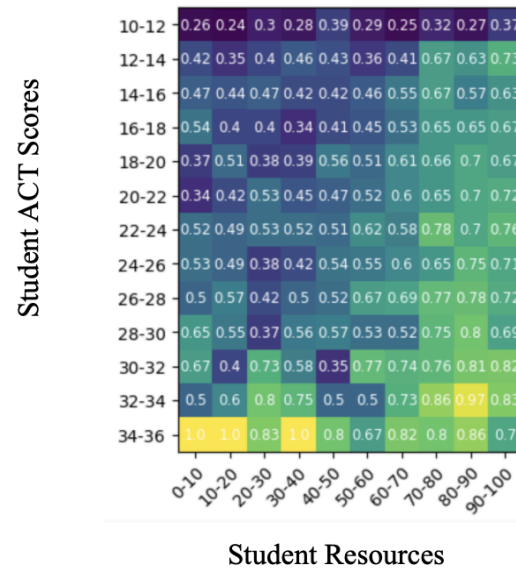
Figure 10: Heatmap of enrollment rate for students by ACT score and student resources (left) and heatmap of enrollment rate for students by essay score and student resources (right) both after 20 years. Comparing student outcomes across resource level in conjunction with their ACT and essay scores shows that wealthier students are more likely to enroll in college. Students with above average ACT scores are also more likely to enroll in college, though this effect is smaller.

Interestingly, there appears to be little difference in student outcomes based on essay score. These disparities in outcomes are much less stark than those in Simulation 2 mostly due to higher enrollment rates outside the quadrant that corresponds to above average ACT scores and above average resources, but a degree of economic inequality still exists.

Acceptance Rate of Students by Resources and ACT Scores



Acceptance Rate of Students by Resources and ACT Scores



Acceptance Rate of Students by Resources and ACT Scores

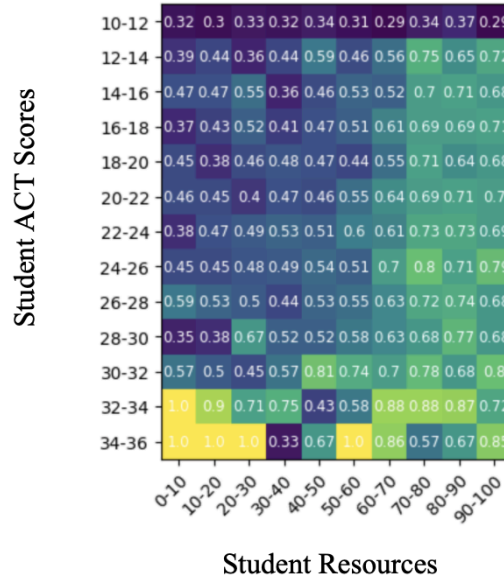


Figure 11: Heatmap of enrollment rates by ACT score and resources for year 1(top left), year 10 (top right), and year 19 (bottom) for the 20 year simulation. Significantly more students were admitted in year 1 than in year 10 or year 19. Furthermore, the heatmaps for year 10 and year 19 show similar trends which indicates that a stable distribution of enrollment rates was reached.

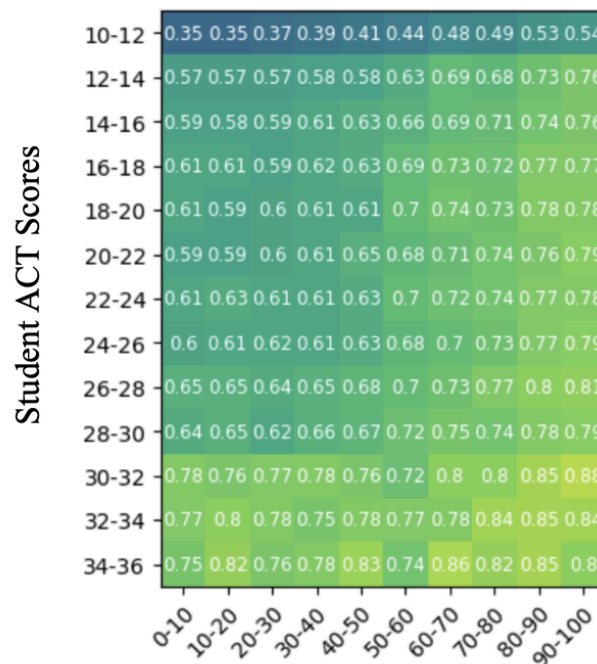
Simulation 4: Remove Wealth Adjusted ACT

Simulation 4 did not adjust ACT scores based on the resource level of students, but included all other mechanisms present in Simulation 3. The below figures summarize the results of the simulation.

	<u>1 year</u>	<u>10 years</u>	<u>20 years</u>
% of students enrolled	71.21	60.06	59.24
Avg resources of enrolled students (/100)	53.21	54.42	54.46
Avg resources of unenrolled students (/100)	44.09	44.26	44.69

Figure 12: Numerical summary of Simulation 4 after 1 year, 10 years, and 20 years. This data is roughly equivalent to the numerical summary of Simulation 3 for each metric.

Acceptance Rate of Students by Resources and ACT



Student Resources

Figure 13: Heatmap of student enrollment rates by ACT scores and student resource level for a 20 year period. The trends shown in the plot are similar to those seen in Figure 10. However, the disparities in enrollment rate across resource level and ACT score are less than in Figure 10. A degree of inequality still exists.

Conclusions

Based on the results of each simulation, in this model the primary driver of economic inequality in higher education was the cost of college. Financial aid softened these disparities in the model, but did not completely eliminate them. The results of Simulation 1 indicate that the model was working correctly because when all mechanisms that create economic inequality are not included, the outcomes are only determined by ACT score as seen in Figures 1 and 2. Comparing the results from Simulations 2 and 3, the inclusion of a financial aid mechanism and an essay score that is not impacted by a student's resource level produces less stark disparities in outcome based on both resource level and ACT score. The heatmaps of acceptance rates by ACT score and resource level and essay score and resource level display this trend. Student attributes that are not impacted by wealth (the essay score in this model) may also play a role in alleviating some of the inequalities. Whether or not the ACT was impacted by a student's resource level didn't produce significant differences in results, but further softened the disparities present in Simulation 3. Comparing the results of Simulations 3 and 4, the outcomes for students show significantly similar trends which suggests that the wealth impact of the ACT did not play as impactful a role in creating the stark disparities seen in Simulation 2 as did the inclusion of cost of college without a financial aid mechanism. However, Simulation 4 shows a further reduction in the differences in outcome across resource level which suggests that it did impact outcomes to a lesser extent than the cost of the college.

Additionally, despite the inclusion of a financial aid mechanism, the results of Simulations 3 and 4 show a positive correlation between the cost of the college and average resource level of the students who enrolled there. This trend is reduced by the inclusion of financial aid by comparing the results of Simulation 2 with those of Simulations 3 and 4, but the positive correlation still exists, especially at colleges with higher costs. This trend makes sense because a college with a high cost needs to have significantly higher financial aid resources to have the same effect on the number of students it can help compared with a college with a lower cost of attendance. The conclusions that were drawn from the results of this model indicate that it was successful in allowing for the examination of the mechanisms of inequality in higher education.

Further Research

Areas for further research and model development include connecting certain attributes of schools together, creating multiple classes of schools that reflect different types of higher education institutions in the United States, and incorporating a mechanism for colleges to compete with each other. The attributes of schools were generated separately from each other and are not tied together (except for the attributes such as acceptance rate that are information that the college keeps). It would be interesting to explore connections between certain attributes such as cost of a college and financial aid resources or cost and minimum ACT. Building upon

that, it would also be interesting to examine the impact of having multiple classes of schools with different ranges for attributes to reflect the different types of colleges in the United States including large private schools (i.e., Stanford), small private liberal arts colleges (i.e., Macalester), state schools (i.e., University of Michigan), and two-year colleges and community colleges (i.e., Southeast Community College in Lincoln, NE). Creating these different classes would allow the model to create colleges that have a set of attributes that reflect a certain type of school. Creating different classes of schools could also open the door to examining admissions data of schools of different types to see if student outcomes are reflected by the results of the model. Finally, in reality colleges can compete with each other for students through certain methods such as merit based aid. Incorporating such a mechanism in the model would help examine how students with high test and essay scores and low resources or students with low test and essay scores and high resources are impacted.

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